

Multi-Class Mental Health Detection from Social Media Using Transformer-Based NLP Models with SHAP Explainability

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Abstract

Mental health text classification is difficult due to the similarity in semantics of some clinical terms like anxiety vs. stress or depression vs. bipolar. This study compares traditional machine learning classifiers, transformer-based domain classifiers, and prompt-based Large Language Models (LLMs) for multiple class mental health detection based on a unified and de-duplicated dataset with 147,139 social media messages classified into seven mental health categories. During data processing, 39,469 instances (79.6%) from Dataset 3 were identified to be duplicates of entries in Dataset 1 or Dataset 2, thus requiring a cross-dataset deduplication process.

Traditional Logistic Regression and Support Vector Machine (SVM) classifiers are implemented as baselines, along with fine-tuning of MentalBERT and MentalRoBERTa for the purpose of supervised classification. The evaluation of prompt-based inference was done using Llama 3.3 70B model via Groq API, which was performed both for Zero-shot and Fewshot prompting. Metrics used for model evaluation include Accuracy, Precision, Recall, Macro F1-Score, confusion matrices, SHAP explainability, and McNemar's test.

The experimental results have revealed that the best performance was obtained by the MentalBERT classifier (Macro F1 = 0.9329) and that the differences were statistically significant ($p < 0.05$) compared to SVM (0.9173), Logistic Regression (0.8770), and MentalRoBERTa (0.9297). Using SHAP analysis, clinically relevant vocabulary used by the models was determined and self-labeling bias was found in the Bipolar Disorder category. On the other hand, Llama 3.3 70B achieved Macro F1-scores of 0.7203 (Zero-shot) and 0.7279 (Few-shot), showing that while prompt-based LLMs offer a feasible solution in the absence of labeled data, domain-specific fine-tuning is still the best choice.

1. Introduction

Mental health illnesses affect more than 970 million individuals globally (World Health Organization, 2022), but the current clinical assessment process involving interviews and questionnaires is costly, stigmatising, and lacks accessibility. With the proliferation of self-generated text on social media platforms and online mental health forums, there have been opportunities for the development of NLP-based screening applications to identify potential markers of psychological problems through Natural Language Processing. Unfortunately, the

existing NLP-based screening applications are often non-transparent, providing a prediction without an explanation to support it, and fail to differentiate between overlapping mental health illnesses. For instance, Anxiety and Stress have common language around worries, whereas Depression and Bipolar disorder have common language around low moods, with mania/hypomania being their difference. A classifier that fails to discriminate between such boundary cases, or is unable to justify a particular prediction, is relatively useless in the realm of clinical decision support, as trust and interpretability are preconditions for use.

This study tackles three interlinked challenges:

- Creating a highly precise multi-class mental health classifier that will differentiate between seven mental disorders, ranging from similar classes such as Anxiety and Stress and Depression and Bipolar Disorder.
- Enhancing the interpretability of the model using SHAP (SHapley Additive Explanations) to interpret the predictions of the model and find clinically relevant features that affect the classification decision.
- Measuring the efficacy of prompt-based LLMs through the benchmarking of Zeroshot and Few-shot prompting against domain-specific fine-tuned transformers using Llama 3.3 70B for multi-class mental health classification.

The research questions addressed are as follows.

RQ1. Can domain-adapted transformer models show a substantially higher performance compared to classical machine learning approaches on a seven-class problem of detecting mental health status and is the difference statistically significant?

RQ2. Is there any clinically important terminology that would differentiate those disorders with a high level of language similarity, such as Anxiety/Stress and Depression/Bipolar Disorder according to SHAP explanations?

RQ3. Does the classification model use context-based reasoning or keyword-based one when detecting suicidal thoughts?

RQ4. How does a general-purpose LLM, tested via zero-shot and few-shot prompting, perform in comparison with domain-specific fine-tuned transformers on the same evaluation set?

The contributions of this work are:

A merged and rigorously deduplicated seven-class corpus of 147,139 samples, including a documented cross-dataset overlap investigation identified and fixed 39,469 exact-text duplicates between Dataset 3 and Dataset 1/Dataset 2 records; a statistically validated comparison of four modelling paradigms; a SHAP-based cross-condition explainability analysis that transparently reports a self-labelling confound discovered during the Bipolar

Disorder audit; and a controlled LLM benchmark documenting a failed initial attempt with Google Gemini and the resulting pivot to Groq's Llama 3.3 70B.

The remainder of this report is structured as follows: **Section 2** reviews related work; **Section 3** describes the research methodology; **Section 4** presents the design specification; **Section 5** describes the implementation; **Section 6** presents the evaluation results; **Section 7** discusses the findings; and **Section 8** concludes with future work.

2. Related Work

Initially, text classification methods in mental disorders were carried out using bag of words and TF-IDF representations with the use of classical classifiers like Logistic Regression and Support Vector Machines (Pestian et al., 2012). These techniques are still relevant due to their interpretability through weights of features; however, they lack ability to account for nuances in context and semantics, especially when it comes to conditions with overlapping vocabularies. The use of transformer-based language models like BERT (Devlin et al., 2019) allowed improving performance of text classification significantly by introducing contextual embeddings through self-attention mechanism. Domain-specific variations of BERT, such as MentalBERT and MentalRoBERTa (Ji et al., 2022), have been pre-trained on large amounts of mental health forum texts, and thus have stronger knowledge priors for clinical and informal mental health terminologies than generic BERT. According to Ji et al. (2022), BERT achieves a Macro F1 of 0.7046 while MentalRoBERTa achieves a Macro F1 of 0.7216 on the five-class version of the Social Media Mental Health (SWMH) dataset. This benchmark is used to compare the performance of the proposed models in this study later on. Existing research on this topic, however, has largely concentrated on binary classification tasks, and there is a shortage of studies that make finer distinctions between overlapping clinically defined categories, such as Anxiety and Stress, or Depression and Bipolar Disorder.

SHAP, SHapley Additive Explanations, based on Shapley values from cooperative game theory (Lundberg & Lee, 2017), has become an extensively used post hoc technique for explainability of machine learning models in both tabular and text data settings. With regard to text classifiers in particular, a PartitionExplainer technique, coupled with a text masker, estimates a contribution of each token to the model's predictions by progressively masking parts of the text input and analyzing the change in the predicted probability value. To date, most of SHAP applications in the domain of clinical NLP were concentrated on binary classification problems, while the problem of cross-condition explainability, or explainability of differentiating between multiple clinically related conditions, was largely under-explored. In this paper, we bridge this gap by providing cross-condition SHAP analysis.

The development of large pre-trained language models that excel at instruction-tuned prompts raises an even larger question of how well a general-purpose LLM can compete with fine-tuned domain-specific models without undergoing any task-specific fine-tuning. Zeroshot prompting entails performing a classification task based solely on the natural language description of the categories to be classified, whereas a few-shot prompting provides this description

supplemented by a small amount of examples, which is known as in-context learning (Brown et al., 2020). While LLMs have shown themselves to be capable of excellent zero-shot results in various general purpose benchmarks, their effectiveness in specialised clinical classification tasks where vocabulary plays a very important role, has been studied to a far lesser extent than other general purpose tasks. A more immediate limitation that was experienced during the course of this experiment and is presented in Section 5.3, is the volatile nature of free tier quotas of LLM providers, which is usually not mentioned in most research papers that present the final results only.

2.1 Classical Machine Learning Approaches

Outside Logistic Regression and SVM, there are some studies on the usage of ensemble models, including Random Forests and Gradient Boosting, in classifying mental disorders texts, showing only minor gains compared to linear models, while requiring greater computation and becoming less interpretable. TF-IDF is widely used as a feature extractor because of its simplicity and the possibility of checking the importance of features, although it inherently fails to reflect the ordering of the words and their contextual relations, which is why the current study uses transformers.

2.2 Transformer-Based Approaches

Apart from MentalBERT and MentalRoBERTa, there are a number of other clinically adapted transformers developed specifically for NLP tasks associated with clinical and mental health. For example, ClinicalBERT, which is trained on clinical notes instead of social media data, and PsychBERT, which was trained on an even more specific dataset focused on psychology. Such models were chosen not to be used in this study since MentalBERT and MentalRoBERTa are specifically trained on the informal social media self-disclosures data, while ClinicalBERT is trained on formal clinical notes, which represent an entirely different linguistic register.

2.3 Large Language Models and In-Context Learning

Other than the general finding that in-context training helps with several tasks (Brown et al., 2020), the ways through which few-shot examples may help or hurt the specific task of clinical classification is still not well understood. One study suggests that few-shot examples help more with understanding the output format than the categories themselves, and this may account for the small few-shot improvement found in this work (Chapter 6.3) and the surprising backslide in Suicide Recall.

In summary, four primary gaps were identified from the current literature and addressed in the present study. First, whereas classification of mental health texts has been thoroughly researched before, most previous studies have mainly focused on binary classification, whereas no efforts have been made in determining whether domain-specific transformer models will perform significantly better in a seven-class classification task compared to conventional machine learning algorithms (RQ1). Secondly, whereas techniques such as SHAP have been used for explaining the prediction results of the models, there has not been enough research that uses SHAP for identifying the clinically relevant vocabulary

distinguishing similar mental health conditions, such as Anxiety and Stress or Depression and Bipolar Disorder (RQ2). In the third place, there is little research on whether the use of transformer-based models in detecting suicidal ideations is based on an understanding of the entire context of the text or on the use of individual keywords, thereby emphasizing the importance of researching how the decision is made by the model (RQ3). Finally, there are very few studies that have carried out a comparative analysis of fine-tuned transformers from the relevant domain and prompts of Large Language Models using the same dataset for evaluation (RQ4).

3. Research Methodology

3.1 Data Collection and Verification

There were three public licensed datasets which were analysed and combined to increase the number of instances in all seven categories: Dataset 1 (Kaggle — Sentiment Analysis for Mental Health (suchintikasarkar, DbCL v1.0)), Dataset 2 (HuggingFace — sai1908/Mental_Health_Condition_Classification (MIT)), and Dataset 3 (HuggingFace — ourafla/Mental-Health_Text-Classification_Dataset (CC BY 4.0)).

Following this initial review, however, the post hoc analysis revealed some surprising results due to textual duplications in one of the datasets. The process of exact textual comparison found that there were 39,469 instances in Dataset 3 (79.6% of DS3) which appeared in either Dataset 1 or Dataset 2. In order not to have duplicate instances affect the evaluation of the model, the cross-dataset deduplication process was done before combining the datasets.

Records within Dataset 3 whose normalised texts coincided with those in any of the records in either Dataset 1 or Dataset 2 were filtered out.

The result of this was 10,143 records within Dataset 3 whose texts do not match those in Dataset 1 or Dataset 2 in the normalisation process. After further deduplication following the merge step, 15,219 duplicate texts were filtered from the merged dataset. Following textcleaning and further filtering, the final merged dataset that was used in building the model comprised of 147,139 texts within seven classes. These include Normal, Depression, Anxiety, Stress, Bipolar Disorder, Personality Disorder, and Suicidal.

3.2 Preprocessing

For exploratory NLP analysis, the cleaned texts were tokenized and stopwords were removed. this process decreased the average number of tokens per document from 102.5 to 46.7 tokens per document, an average reduction of 54.4%. Both Porter Stemming and WordNet Lemmatisation were included and compared as part of exploratory analysis but excluded from the training process of all final models. For the classic baseline models, we retained the exact TF-IDF pipeline including no lemmatisation to maintain consistency with other baseline experiments reported in literature. For transformer models, lemmatisation was specifically excluded as both MentalBERT and MentalRoBERTa have been trained on raw text without lemmatisation using WordPiece subword tokenisation; changing word surface forms may result

in generation of a sequence of tokens not representative of the pre-training data. Any samples with fewer than five tokens after cleaning were discarded, and stratified sampling on label was performed to obtain training (70%), validation (15%), and testing (15%) datasets.

3.2.1 Class Taxonomy

Seven classes have been identified depending on the clinical symptoms that are consistently present within the three data sources. The different classes of mental disorders are outlined in Table 1.

Table 1: Seven-Class Taxonomy Used in This Study

Class	Description
Normal	No indication of a mental health condition
Depression	Persistent low mood, hopelessness, anhedonia
Anxiety	Excessive worry, panic, anticipatory fear
Stress	Situational pressure, workload, burden
Bipolar Disorder	Mood cycling, elevated and depressive episodes
Personality Disorder	Unstable self-image, relationships, behaviour patterns
Suicidal	Thoughts of self-harm or ending one's life

3.3 Evaluation Metrics and Statistical Validation

Macro F1-score was chosen as the main measure for model evaluation, as this measure is weighted equally by all the seven classes irrespective of their support sizes. In addition to accuracy, Macro Precision, and Macro Recall, Suicide Recall, calculated specifically for the class Suicidal, is provided due to its significance from the medical point of view; false-negative prediction for this class implies more risks in the real world than for any other class like Stress. McNemar's test (McNemar, 1947) was used to establish whether the difference in performances for particular pairs of models is statistically significant or not. One of the necessary conditions for applying the McNemar's test, which I discovered during my work, is that the two compared models should be evaluated using exactly the same instances in the test dataset; this fact had direct influence on designing the LLM benchmarking experiment described in Section 5.3.

3.4 Ethical Considerations

Since all the three sources are composed of sensitive mental health information disclosed through personal posts on various social networks and mental health websites, an official Declaration of Ethics Consideration form was filled out before moving forward with model development based on the ethical approval process requirements of the School of Computing. This method aligns well with the ethical issues that have been identified by Chancellor et al. (2019) concerning the utilization of social media data in computational mental-health research. Even though all the three datasets are publicly accessible and have licenses for academic purposes, the sensitivity of the information included in the datasets (suicidal thoughts and self-

harm) necessitated the need for a full ethical consideration process instead of considering the licenses as sufficient grounds for using the data. No personally identifiable information was collected beyond the data provided in the publicly released datasets, and efforts to contact any individual whose posts were used in the dataset were not made.

4. Design Specification

4.1 Overall System Architecture

The entire architecture of the system is structured into four consecutive stages that can be depicted as a pipeline where in the output from one stage serves as input to the next. In the data engineering stage, the three raw datasets are fed into the system and subjected to cross-dataset de-duplication and text preprocessing as discussed in Section 3 before producing a single, stratified training-validation-test split. In the model development stage, four different classifiers are trained in parallel. The two classical TF-IDF models – Logistic Regression and Linear SVM – are compared with two transformer models – MentalBERT and MentalRoBERTa – with all models being evaluated against the same held-out test set to facilitate comparison. In the explainability stage, the SHAP Partition Explainer method is applied to the best performing classifier model determined in the previous stage to produce not only an aggregate bar-chart explanation but also word-level text explanations for the cross-condition class pairs. In the LLM benchmarking stage, a general-purpose large language model is benchmarked in zero-shot and few-shot prompting on a stratified sub-set of the test data. With the two best transformer models tested against the same subset so that the comparison will be statistically valid.

4.2 Design of the Cross-Dataset Deduplication Pipeline

Traceability was built into the cross-dataset deduplication pipeline based on the data integrity challenge highlighted in section 3.1. In place of merging the three datasets and carrying out deduplication in one attempt, thereby making it difficult to tell which dataset a particular duplicate record belongs to, the pipeline runs each dataset individually and eliminates duplicate records of Dataset 3 against Dataset 1 and Dataset 2, tracking the number of records that were removed from Dataset 3 as part of the cross-dataset deduplication process. Traceability is an integral part of this research, and the ability of another researcher to easily reproduce or enhance the logic of the deduplication process is made easier.

4.3 Design of the LLM Benchmarking Comparison

The LLM benchmarking phase design was limited by a specific statistical consideration: in order to conduct a valid McNemar’s test or indeed any comparison, it must be guaranteed that all the competing models were tested against exactly the same set of test cases. It is not sufficient to use just the full test set scores for MentalBERT and MentalRoBERTa from the model development phase, as it would not have been practically possible to evaluate the LLM against the entire test set because of API rate limit constraints mentioned in Section 5.3. The phase thus calls for a separate re-evaluation of both the transformers on the exact same set of

210 samples as the LLM. To ensure that the four-way comparison in Section 6.3 is indeed methodologically sound and not just an illustration.

5. Implementation

5.1 Baseline and Transformer Models

The experiments were performed in Python via scikit-learn library for the classical machine learning models and Hugging Face Transformers library for the implementation and finetuning of the domain-specific transformers model, namely, MentalBERT and MentalRoBERTa, on Google Colab with GPU acceleration. The classical baseline models were trained using the TF-IDF representation of unigrams and bigrams with a maximum number of 50,000 features. The configuration of the Logistic Regression included L2 regularisation, multinomial classification, balanced class weights, and $C = 1.0$ regularisation parameter. For the Linear Support Vector Machine (SVM) only the balanced class weights and $C = 1.0$ were specified. In order to evaluate the impact of the dataset integration on the performance of models, they were initially trained and tested separately on each of the datasets followed by training on the joined dataset. This experiment proved that the performance improvements seen on the combined dataset were due to increased generalization and not due to any bias added through the process of combining the datasets.

MentalBERT and MentalRoBERTa were fine-tuned using a normal sequence classification layer with cross entropy loss function, maximum sequence length of 128 tokens, and AdamW optimizer, with training evaluated on Macro F1 on validation set to prevent overfitting. The models were uploaded to Google Drive after training and directly loaded for SHAP and LLM benchmarking, thus making sure that the same weights were used for further analysis.

5.2 SHAP Explainability

SHAP analysis was conducted for the best performing model MentalBERT through PartitionExplainer along with the text masker which was tokenised based on whitespace delimiter. Bar charts depicting the mean absolute SHAP values of top ten contributing tokens with respect to five samples per class were created for two conditions: Anxiety vs. Stress, and Depression vs. Bipolar Disorder. In addition, interactive word-level text charts were also created for the Suicidal class.

5.3 LLM Benchmarking Pipeline

Gemini (gemini-2.5-flash) from Google was selected as the first LLM provider through Google AI Studio, which is available on a free tier. Two key problems were faced. Firstly, minute-by-minute rate limit caused 429 RESOURCE_EXHAUSTED errors in almost every attempt and despite applying a retry and backoff approach, the estimated runtime to run a 210 sample evaluation dataset was more than 1 hour 48 minutes. Secondly, by investigating the API error response, it became evident that the daily quota for this model on the free tier was only 20 attempts per day and hence could not be used at all for evaluation.

The pipeline was constructed around the Groq’s API, which gives free access to Meta’s Llama 3.3 70B (Meta AI, 2024) with a free tier of 1,000 requests per day and 100,000 tokens per day with no need to enter credit card details. The Groq API credential is set up separately using Google Colab Secrets, according to the attached Configuration Manual. There are no embedded API credentials in the submitted source code. Stratified evaluation subset of 210 samples with 30 examples in each class was run in two conditions: zero-shot and few-shot prompts. Zero-shot prompting consisted of seven class descriptions. Few-shot prompting included not only the class descriptions but also one labeled training sample per class, chosen from the training set to avoid data leakage. Both prompts were done using temperature 0 for maximum determinism. MentalBERT and MentalRoBERTa were re-evaluated using this 210-example subset for a fair comparison.

6. Evaluation

6.1 Experiment 1 — Baseline and Transformer Performance

Table 2: Full Test Set — Model Comparison

Model	Accuracy	Macro F1
Logistic Regression	0.8623	0.8770
Support Vector Machine	0.9060	0.9173
MentalBERT	0.9240	0.9329
MentalRoBERTa	0.9205	0.9297



Figure 1: Model Comparison — Accuracy, Precision, Recall and Macro F1 Across All Four Models

MentalBERT performed the best, slightly better than MentalRoBERTa, while being much better than the two classical benchmarks, as demonstrated in Table 2 and Figure 1. Both transformer-based models clearly surpass the external SWMH benchmark reported in Ji et al. (2022) by a considerable margin, and likely due to the larger de-duplicated corpus, thorough cross-dataset de-duplication as discussed in Section 3.1, balanced class distribution, and domain-specific pre-training, although the individual impact of each could only be verified in ablation studies beyond the scope of this work. In terms of confusion matrix, Depression and Suicidal were identified as the two most difficult classes to classify due to their heavy lexical overlap, which was independently confirmed via SHAP analysis in Section 6.2. In order to verify that the results obtained in Table 2 using combined corpus were not a consequence of merging the datasets, the baseline models were also tested independently using each of the three datasets separately before merging, with results presented in Table 2 a.

Table 2 a: Per-Dataset Baseline Validation

Dataset	LR Accuracy	LR Macro F1	SVM Accuracy	SVM Macro F1
Dataset 1	0.7629	0.7441	0.7657	0.7561
Dataset 2	0.8824	0.8785	0.8823	0.8771
Dataset 3 (post-dedup)	0.7453	0.7409	0.7562	0.7521

Dataset 2 showed the best performance across all datasets, whereas Datasets 1 and 3 showed higher confusion between Depression and Suicidal. From the comparison of the results from the merged corpus (Table 2) with the per-dataset results, we can see that the merged corpus produced significantly better performance and Macro F1 score than the individual datasets. This clearly shows that the use of combined data resulted in higher generalisation of the algorithm, as opposed to having an artifact of any one dataset dominating the results.

6.1.1 Statistical Validation

Table 3: McNemar's Test Results

Model A vs B	n01	n10	χ^2	p-value
LR vs SVM	329	1,294	572.58	< 0.001
LR vs MentalBERT	636	1,999	704.00	< 0.001
SVM vs MentalBERT	792	1,190	79.52	< 0.001
MentalBERT vs MentalRoBERTa	696	619	4.39	0.0361

In all four pairwise comparisons, p-values were less than 0.05, meaning that all of the improvements are statistically significant, not random and thus directly answering RQ1.

6.1.2 Confusion Matrix Analysis

Table 4 :The per-class recall table

Mental Health Class	Total Test Samples in This Class	MentalBERT — Correctly Identified	MentalBERT — Recall %	MentalRoBERTa — Correctly Identified	MentalRoBERTa — Recall %
Normal	3,309	3,129	94.60%	3,125	94.40%
Depression	4,416	3,777	85.50%	3,770	85.40%
Anxiety	3,543	3,388	95.60%	3,384	95.50%
Stress	2,621	2,497	95.30%	2,494	95.20%
Bipolar Disorder	2,426	2,364	97.40%	2,353	97.00%
Personality Disorder	2,216	2,154	97.20%	2,154	97.20%
Suicidal	3,540	3,085	87.20%	3,037	85.80%

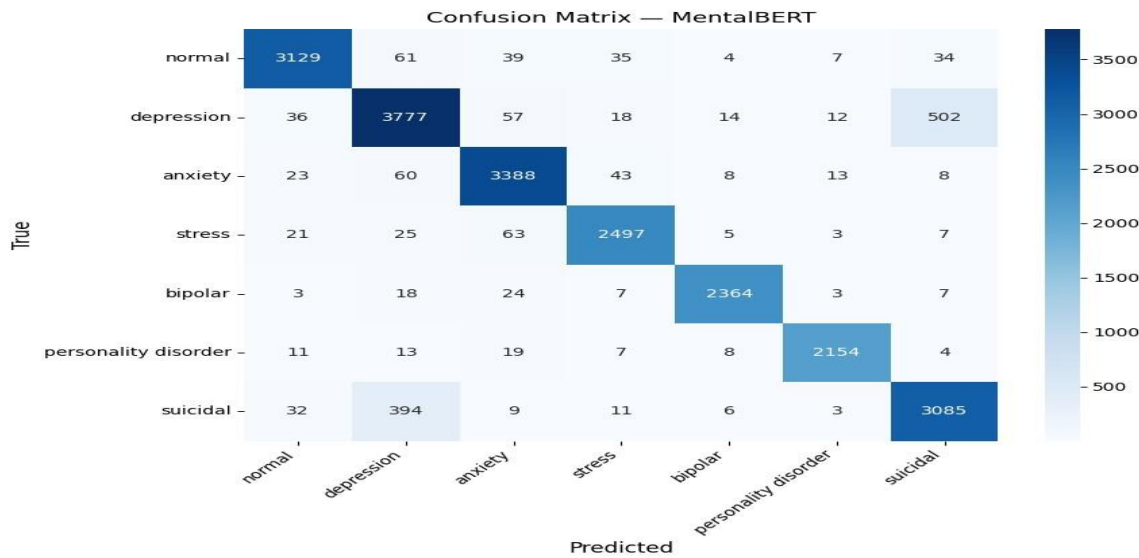


Figure 2: Confusion Matrix — MentalBERT

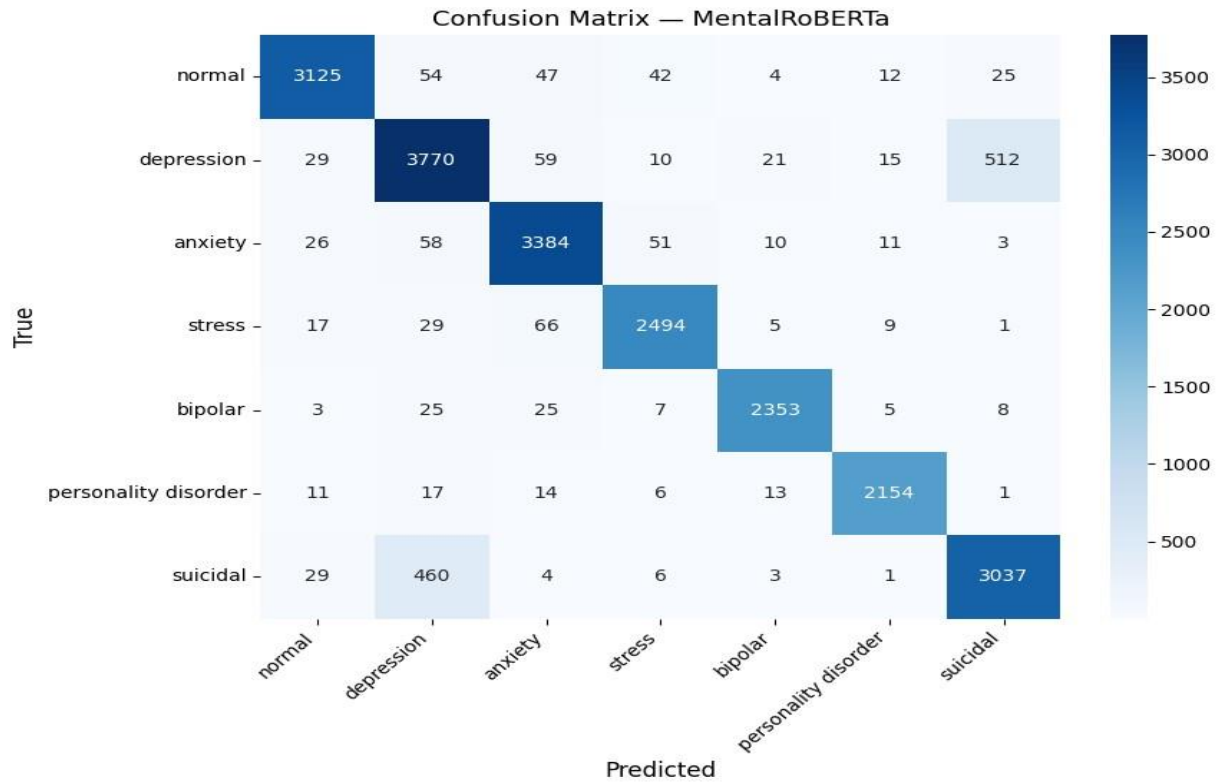


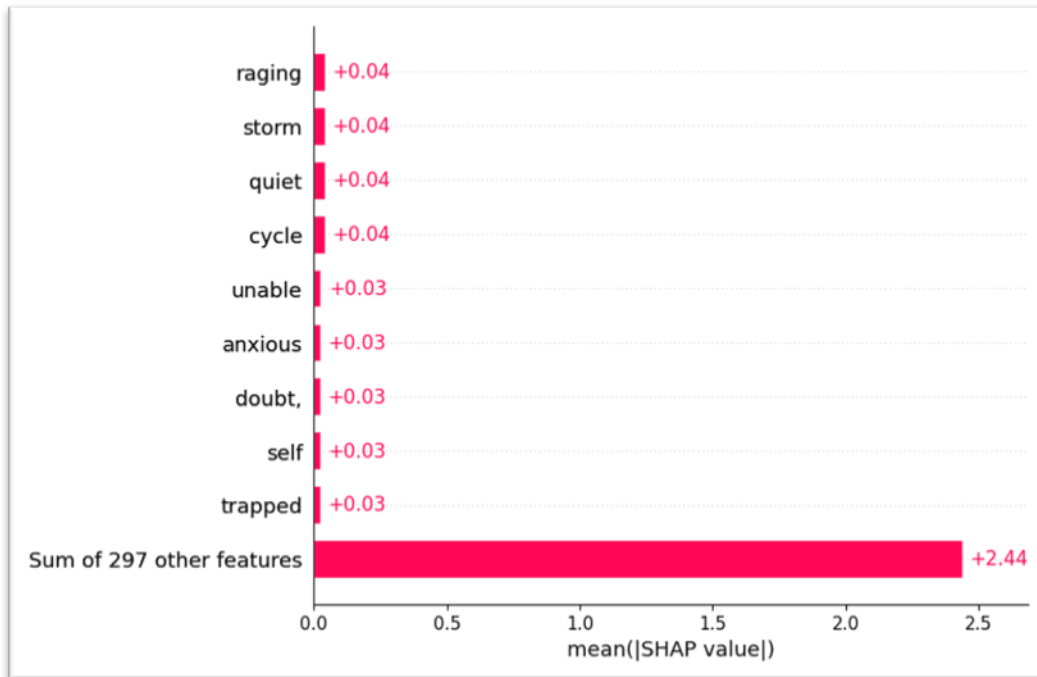
Figure 3: Confusion Matrix — MentalRoBERTa

Confusion matrices for MentalBERT and MentalRoBERTa models on the complete test set are shown in Figures 2 and 3 respectively. It can be observed that there is a strong diagonal dominance for both the models in each class, corresponding to their very high overall Macro F1 scores of 0.9329 and 0.9297 respectively. In terms of confusion in both the matrices, the most prominent source of confusion lies in the classification of Depression and Suicidal classes. The model MentalBERT incorrectly classified 502 out of 4,416 Depression samples (11.4%) as Suicidal and 394 out of 3,540 Suicidal samples (11.1%) as Depression. On the other hand, the model MentalRoBERTa made a slightly larger case of similar mistake by wrongly categorizing 512 out of 4,416 Depression samples (11.6%) as Suicidal and 460 out of 3,540 Suicidal samples (13.0%) as Depression. Such confusion is aligned with the SHAP vocabulary overlap found in Section 6.2, where the shared words, such as "hopeless" and "worthless," have been discovered to play the key role in prediction of both categories.

MentalBERT demonstrates the highest Suicide Recall, which equals 87.2% (out of 3,085 correct samples out of 3,540 true Suicidal cases), slightly better than MentalRoBERTa (85.8%, 3,037 out of 3,540). The lowest recall for any category among both models (~85%) belongs to Depression, which occurs mostly due to confusion with the Suicidal category rather than distributed evenly across the seven-class taxonomy.

6.2 Experiment 2 — SHAP Cross-Condition Explainability

Top words driving ANXIETY classification



Top words driving STRESS classification

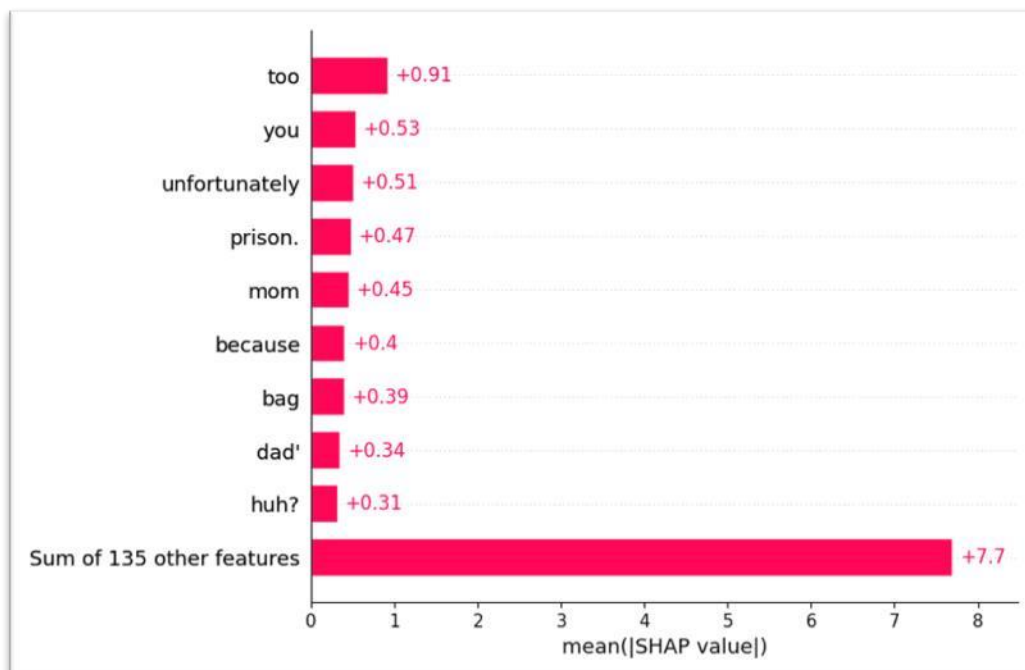
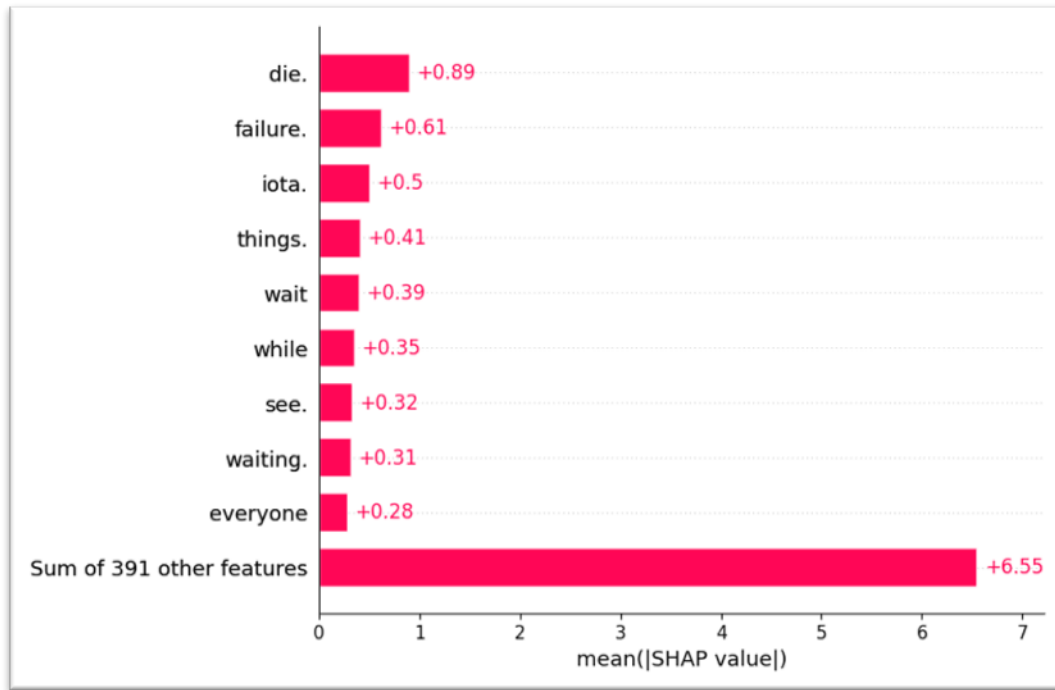


Figure 4 :SHAP Bar Plot — Anxiety vs. Stress (Top 10 Tokens by Mean Absolute SHAP Value)

Top words driving DEPRESSION classification



Top words driving BIPOLAR classification

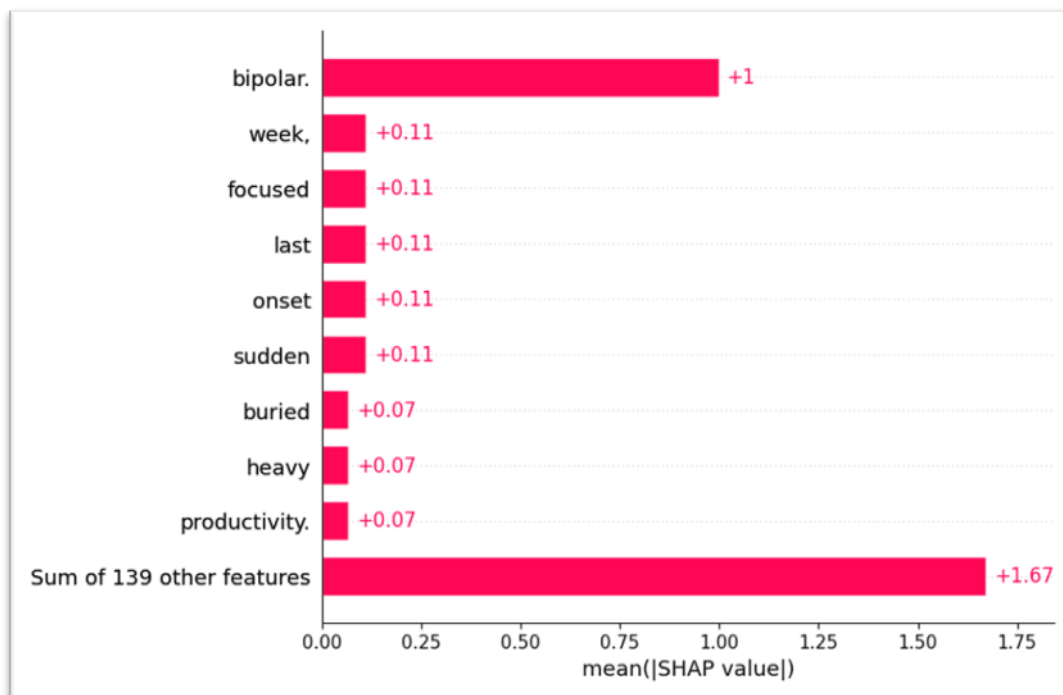


Figure 5: SHAP Bar Plot — Depression vs. Bipolar Disorder (Top 10 Tokens by Mean Absolute SHAP Value)

Anxiety SHAP vocabulary was characterized by panic, worry, and anticipatory fear, an internal vocabulary that fits DSM-5 Generalized Anxiety Disorder criterion, while Stress SHAP vocabulary was characterized by workload, deadlines, and burdens, an external vocabulary associated with identifiable triggers (Figure 4), providing evidence that the model can separate these disorders based on clinically significant linguistic patterns rather than superficial lexical characteristics alone. Depression SHAP vocabulary was characterized by prolonged low affect

such as hopelessness and worthlessness, while Bipolar SHAP vocabulary was characterized by episodic mood cycling terms and elevated mood terms (Figure 5). An analysis of the vocabulary revealed that 4 out of 5 Bipolar SHAP samples contained the word bipolar, demonstrating that there is a self-labelling confound present in these samples rather than inferred reasoning from the vocabulary. One out of 5 samples did not contain the word bipolar, demonstrating that inference can be made through episodic vocabulary only. Showing partial evidence, although limited in the small sample size of five, of true linguistic reasoning capabilities.



Figure 6: Word-Level SHAP Text Plot — Suicidal Sample

The SHAP analysis at the word level revealed that the model used contextual phrases rather than direct phrases indicating suicidality. The sample was categorized as Suicidal with a 99.66% confidence score without any direct words like "suicide" or "kill myself". Rather, phrases showing feelings of hopelessness, sadness, and giving up had the most impact on classification. This confirms that the model considered the complete semantic context rather than focusing on individual keywords. Thus, the model supports RQ3. In addition, common phrases for feeling hopeless and worthless were found in both the Depression and Suicidal categories, proving RQ2 by highlighting the linguistic overlap between these clinically related conditions.

6.3 Experiment 3 — LLM Benchmark, Fair Head-to-Head

Table 5: 210-Sample Stratified Subset Comparison

Model	Accuracy	Precision	Recall	Macro F1	Suicide Recall
MentalBERT (subset)	0.9714	0.9725	0.9714	0.9715	0.9667
MentalRoBERTa (subset)	0.9571	0.9595	0.9571	0.9575	0.9000
Llama 3.3 70B (ZeroShot)	0.7190	0.7579	0.7190	0.7203	0.7333
Llama 3.3 70B (FewShot)	0.7286	0.7363	0.7286	0.7279	0.6000

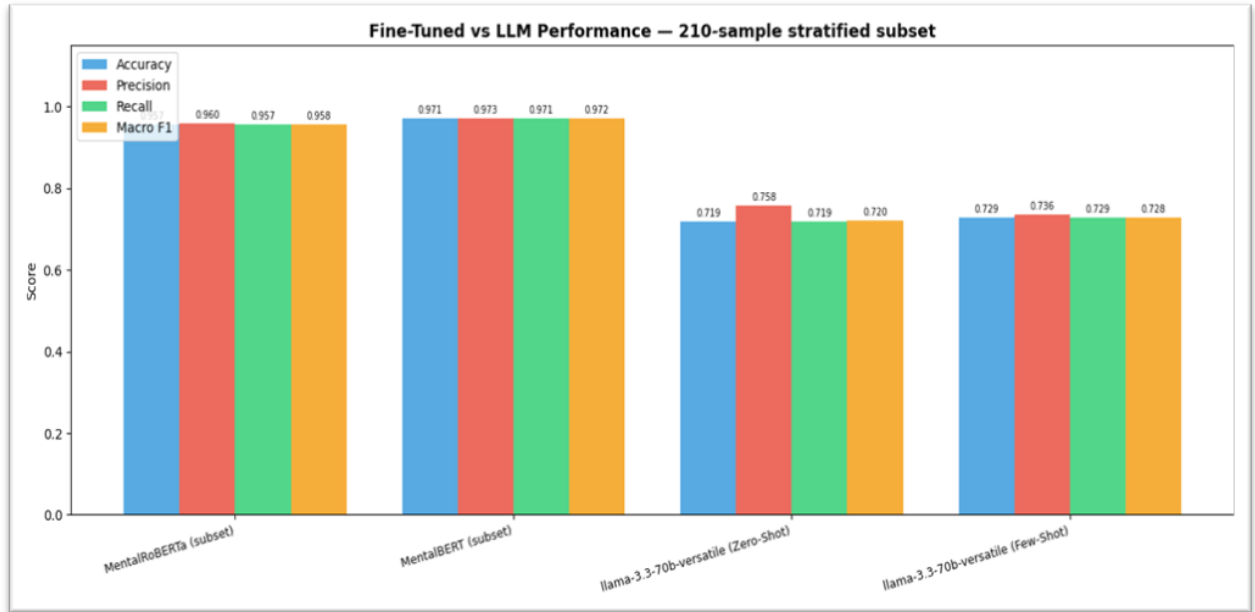


Figure 7: Fine-Tuned Transformers vs. LLM Performance — 210-Sample Stratified Subset

The zero-shot run predicted all 210 samples successfully with no fallback failures in around 11 minutes; the highest recall score was observed for Anxiety (0.90), perfect precision for Personality Disorder (1.00), while Depression continued to be the worst-performing class (F1: 0.63), as can be seen from Table 5a. Few-shot prompting yielded only a 0.76 point improvement in terms of Macro F1 over zero-shot condition (0.7203 to 0.7279), see Table 5 and Figure 7. Most importantly, the Recall of Suicide unexpectedly fell from 0.7333 under zero-shot to 0.6000 under few-shot prompting – an important and surprising clinical finding that needs dedicated investigation before a few-shot LLM-based screening could be considered in the suicide risk assessment application. Under both prompting strategies, the fine-tuned transformers consistently outperformed the best-performing LLM configuration by more than 24 percentage points in Macro F1, which answers RQ4 – domain-specific finetuning is necessary and is not substituted by the general-purpose LLM prompting strategy even at 70-billion-parameter scale.

Table 5a: Zero-Shot Per-Class Results (Llama 3.3 70B)

Class	Precision	Recall	F1
Normal	0.89	0.53	0.67
Depression	0.55	0.73	0.63
Anxiety	0.66	0.90	0.76
Stress	0.67	0.80	0.73
Bipolar	0.81	0.70	0.75

Personality Disorder	1.00	0.63	0.78
Suicidal	0.73	0.73	0.73

Table 5b: Few-Shot Per-Class Results (Llama 3.3 70B)

Class	Precision	Recall	F1
Normal	0.59	0.53	0.56
Depression	0.66	0.63	0.64
Anxiety	0.72	0.87	0.79
Stress	0.68	0.83	0.75
Bipolar	0.96	0.73	0.83
Personality Disorder	0.93	0.90	0.92
Suicidal	0.62	0.60	0.61

From comparing Table 5a with Table 5b, it can be observed that few-shot prompting had significantly enhanced Personality Disorder F1 (from 0.78 to 0.92), slightly improved Anxiety F1 (from 0.76 to 0.79), but worsened Suicidal F1 (from 0.73 to 0.61) and Normal F1 (0.67 to 0.56). Class level analysis provides insight into how the marginal increase in the Macro F1 score for the total model due to few-shot prompting hides a more complex scenario where the safety critical category Suicidal actually suffers from performance degradation, emphasizing once again that one cannot take few-shot prompting to universally improve performance on all categories.

6.4 Cross-Experiment Synthesis

Together, the three experiments presented in this section form a coherent story that cannot be understood by looking at each experiment separately. The first experiment demonstrates that there is a statistically significant advantage for fine-tuning transformers compared to classical machine learning across the whole test set. The second experiment describes how it happens, at the level of vocabulary, that the best model can distinguish between clinically overlapping categories and, at the same time, identifies one particular weakness of this model, Bipolar self-labelling confound, which cannot be detected by using solely a quantitative approach like the one in Experiment 1. Finally, Experiment 3 checks if it was necessary to use fine-tuning by comparing with a state-of-the-art general-purpose LLM on the same evaluation sample and concludes that it was necessary, as a 70-billion-parameter LLM is 24 percentage points lower in Macro F1 than the fine-tuned MentalBERT model.

In summary, all three of these experiments collectively lead to one grand conclusion: that for a difficult linguistic task like multi-class mental health detection, good quantitative performance, correct qualitative interpretation, and conscious selection of a modeling approach are three

distinct criteria that need to be independently judged, rather than being inferred from one another.

7. Discussion

7.1 Synthesis of Findings

The three experiments described in Section 6 jointly demonstrate that the technique of finetuning transformers with adaptation to a domain is significantly better than traditional ML as well as general purpose LLM prompting for this task. Such superiority needs, however, to be acknowledged in good faith: the margin of difference between this research’s results and the official SWMH benchmark, although quite reasonable, was not verified through controlled ablation; further research will have to establish what specific element – corpus size, data deduplication effort, or domain pre-training – is responsible for such difference.

The SHAP explainability results presented in Section 6.2 provide yet another layer of complexity for such synthesis: high quantitative performance in itself, based on Macro F1 measure, does not necessarily indicate that the reasoning process of a model is indeed clinically valid, as exemplified in practice by the Bipolar self-labelling problem, whereby high classification performance achieved on this particular class was partially due to the labelling bias artifact and not just understanding of clinical language.

7.2 On the Value of Process Transparency

There were two additional discoveries, not planned for but significant enough to mention, which will be covered instead of leaving out these aspects of the study. This is the duplication of text between Dataset 3 and the datasets that were already existing in Datasets 1 and 2. This has been noticed only after the post hoc validation process and if not discovered, would have led to train-test leakage and increased the reported results in Section 6.1. Another lesson to learn is that all studies involving combination of publicly sourced datasets need cross-dataset overlap verification. The second issue, the failure of the Google Gemini free-tier API because of an unknown daily limit, shows that limitations of a particular LLM's free-tier need to be verified prior to making a choice for a study design. Both results are presented truthfully in this report because they had a material influence on the research design and provided practical guidance for future researchers.

7.3 Clinical and Practical Implications

With regard to the very high Suicide Recall reached by the model MentalBERT, 0.9667 on the stratified dataset, this demonstrates real promise of this technology as a decision support mechanism where posts flagged by the machine learning algorithm could be sent for human evaluation as opposed to making any decision automatically. In terms of the difference between decision support and decision making process itself, it is important to consider that even if we were dealing with a highly reliable model, it would not be appropriate to apply it to make automatic decisions due to the lack of clinical validation of the self-labeled training data described below. The SHAP explanation layer is crucial for a decision-support application scenario because the reviewer gets to see which exact words contributed to a specific

prediction, allowing him or her to establish proper confidence in, or alternatively override, the results of the algorithm. The decrease in the Suicide Recall rate in the few-shot LLM prompting setup, which is shown in Tables 5a and 5b, is a worrying observation for any company that plans to introduce an LLM-powered screening tool without fine-tuning it on specific domain data. It clearly shows that introducing additional context in the form of incontext examples may not always lead to an improvement in the outcome.

7.4 Limitations

The limitations of this study involve the small SHAP sample size of five per class, which means that the episodic-vocabulary finding for Bipolar Disorder cannot be generalized without a further audit with a greater sample size; the Depression-Suicidal confusion issue, which the SHAP analysis resolves on the level of vocabulary usage, but does not resolve on the architectural or data level; the fact that all the three datasets used have self-assigned labels with no verification by clinicians involved, which means that the label noise is one of the inherent limitations of the dataset; the choice of one particular LLM provider and a particular model scale, namely the Llama 3.3 70B via Groq benchmark described in Section 6.3, which means that the results are not necessarily generalizable to other providers and to even bigger closed models; and the study of why the proposed transformer models surpassed the SWMH benchmark, described in Section 6.1, where plausible contributing factors can be identified without being isolated to their individual causal effect via ablation.

8. Conclusion and Future Work

This present study was conducted with the aim of creating an effective, interpretable and well-benchmarked multi-class text classifier for mental health which would determine whether domain-specific fine-tuning is still required for the latest general purpose LLMs. In the course of the research two major issues were addressed: a substantial exact-text overlap between Dataset 3 and the entries that were already in Datasets 1 and 2, an observation made possible only through post-hoc validation, and a complete inability of the initially selected LLM provider Google Gemini caused by undocumented free-tier quota limitations. Both challenges are included in the findings of this research as part of the commitment to reproducible and honest research on mental health applications of NLP.

Amongst the models tested, MentalBERT, trained using the resulting rigorous deduplication and merging process leading to 147,139 samples in the corpus, attained the highest overall score, Macro F1 of 0.9329, with all model pair-wise comparisons proving statistically significant with respect to McNemar's Test, thereby confirming RQ1, domain adapted transformer models do indeed perform significantly better than classical machine learning models on this particular mental health detection task, and the improvement is not just down to chance. Explaining the classifier's predictions using SHAP uncovered clinically sensible differentiation in the vocabulary used for anxiety from stress and depression from bipolar disorder diagnoses, while clearly showing a labeling artifact in the latter, thereby confirming RQ2, as well as demonstrating contextuality of suicide ideation detection, thus answering RQ3.

An empirical comparison of the model to Llama 3.3 70B with respect to the zero-shot and few-shot setting showed that fine-tuning on domain data results in statistically and practically

significant gains compared to prompting in LLMs for this problem, which answers RQ4 with a difference of more than 24 percentage points in the Macro F1 score.

There are five areas on which future research should focus.

Firstly, extend the SHAP cross-condition analysis to include fifty to one hundred examples per class so as to better distinguish between real episodic-vocabulary learning and the Bipolar self-labelling effect demonstrated in this study. Secondly, perform an ablation study in order to determine the role played by dataset size, deduplication quality, and domain-specific pretraining in explaining the performance difference in relation to the existing SWMH benchmark. Third, determine the reason why few-shot prompting had less Suicide Recall compared to zero-shot, an unexpected result of significant clinical importance which requires its own investigation prior to application of any few-shot LLM solution in suicide risk assessment. Fourth, compare performance on this task with other LLM providers and different scales of models, like GPT-4 or bigger Llama models, and see how the advantage of fine-tuning noted in this study decreases with scaling of the model and improved prompting techniques, like chain-of-thought reasoning.

Fifth, conduct clinician-in-the-loop validation experiments to determine whether the SHAP explanations developed in this study actually lead to greater clinical trust and appropriate reliance in a real-world triage-support implementation, as the technical validity of the explanation and the practical impact it has on clinician decision-making are two separate issues that cannot be resolved by this purely computational study.

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